

Grades for Erneste Ntezirizaza

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Course

Mission Capstone

Arrange By

Due date

Apply

Name	Due	Submitted	Status	Score	
Honour code Assignments	25 May by 23:59	12 May at 9:34		1 / 1	
Initial software product/solution demonstration Assignment 2: Initial Design and Development [F:A]	4 Jun by 23:59		missing	0 / 0	
Assignment 1: Research Proposal Assignment 1: Research proposal [F:A]	9 Jun by 23:59	9 Jun at 23:39		15.6 / 19	  

Assessment by Dirac Murairi

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1. Project Proposal

Criteria	Ratings				Points
Research Topic, Problem Statement, and Questions	2 to >1.5 pts Mastery Clearly defined, original, relevant problem with focused research questions. Comments The problem statement is clearly organized into three distinct, well-evidenced problems, each backed by a specific figure (0–20% survival	1.5 to >1 pts Proficient Relevant topic but lacking depth or clarity in some aspects.	1 to >0 pts Proving proficient Problem statement and questions need refinement.	0 pts Fail Missing or poorly defined problem and questions.	1.5 / 2 pts

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	rate, 2% annual Eastern Province loss, the synthetic-only limitation of Siedler 2022), and the topic itself is relevant and grounded in real, current data rather than an assumed need. The research questions are specific and testable: RQ2 and RQ3 in particular map directly onto planned ablation conditions (reward components, curriculum learning), which means they can actually be answered by the methodology rather than sitting as aspirational framing. The recurring issue is that the problem statement and the first research question don't stay neutral about the solution. The problem statement closes by naming the specific technical choice directly, "this work introduces ARIA, a PPO-based reinforcement learning system... using a custom Gymnasium environment," and RQ1 asks "to what extent does ARIA's PPO agent outperform random, uniform-grid, and greedy baselines," which bakes the algorithm choice into the question itself rather than asking a more open question about whether a trained RL agent outperforms the baselines. A problem statement should be able to stand on its own without naming the tool that will fix it, and a research question should be answerable in a way that doesn't presuppose which specific algorithm was the right one to use. As written, RQ1 can only really be answered as "yes, PPO does" or "no, it doesn't," since PPO is already fixed inside the question rather than being one candidate under test. The fix is straightforward: keep PPO's justification in the methodology chapter where it belongs, and rewrite the problem statement's final sentence and RQ1 to refer to "a reinforcement learning agent" rather than naming PPO specifically, which would let the research question actually interrogate the choice of learning approach rather than assume it.				
Background and Context	2 to >1.5 pts Mastery Comprehensive overview with relevant literature and context.	1.5 to >1 pts Proficient Adequate overview but lacks depth.	1 to >0 pts Developing Incomplete discussion of literature and context.	0 pts Fail Insufficient background information.	1.5 / 2 pts
	Comments The background section grounds itself in real, verifiable data rather than vague claims, and this is its main strength. The figures on 2024 forest loss (910 hectares, roughly 500 kilotons of CO2 equivalent) match Global Forest Watch's country dashboard for Rwanda exactly, and the 10.7% to 30.4% forest cover growth figure between 2010 and 2022 is correctly attributed to the IISD report on forest conservation in producing countries. The AirSeed Technologies figure of 40,000 seed pods per day also checks out against the cited Mongabay article. This kind of accurate sourcing is exactly what this section should do, and it gives the reader confidence that the scale of the problem being described is real rather than asserted. Where the section falls short of full marks is in how narrow the framing becomes once the statistics are laid out. After establishing the deforestation numbers, the section moves almost immediately into				

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	naming specific commercial drone systems and their throughput figures, which is material that belongs to and is later repeated in the literature review chapter, rather than using this space to build out the broader context: the socio-economic drivers of deforestation beyond a single sentence, the policy history behind Rwanda's forest cover recovery, or how the ecological stakes connect to livelihoods and climate goals beyond NST2. The section also starts introducing the RL solution and its justification quite early, which crowds out space that could deepen the contextual picture before the problem statement takes over that job. Broadening the discussion of context and drivers, and holding back the AirSeed/MORFO detail for the literature review where it is examined properly, would move this into the mastery band.				
Clear research gap	2 to >1.5 pts Excellent Clearly identifies and justifies the gap using recent literature or societal needs.	1.5 to >1 pts Good Gap identified but lacks supporting evidence.	1 to >0 pts Fair Problem discussed but gap unclear.	0 pts Poor No identifiable research gap.	1.5 / 2 pts
Comments The proposal correctly identifies a narrow, defensible gap: no published system trains a reinforcement learning agent on real African geospatial data for aerial seed placement, and Siedler (2022) is accurately characterised as the closest prior work with a synthetic, non-ecological environment. That specific claim survives scrutiny against recent literature. The broader claim in the problem statement, however, that no deployed aerial seeding system anywhere performs adaptive, in-flight decision-making, is contradicted by a 2026 vision-guided UAV system that adjusts seed dispensing in real time based on onboard terrain classification. That system does not use reinforcement learning, so it does not close the narrower gap ARIA targets, but its existence means the stronger, sweeping version of the claim should be softened. There is a second, more structural issue with how the gap is framed. The gap is essentially a "no one has combined X and Y" gap: reinforcement learning has not been combined with real Rwanda-specific geospatial data. That is a legitimate kind of gap, but the proposal never argues why closing it with reinforcement learning specifically, rather than a classical constrained-optimization or heuristic approach applied to the same real data, is the right way to fill it. Given that the environment is static, fully observable, and modest in size, a non-learning method could plausibly reach similar performance, and the proposal's own expected-results section anticipates only a small margin over the greedy baseline, which quietly supports that concern. The gap would be considerably stronger if it named generalization to unseen terrain zones as the reason the					

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Objectives view longer description	combination matters, since that is the one property a trained policy has that a re-optimized solution does not, rather than resting the justification on novelty of combination alone.				
	1 to >0.75 pts	0.75 to >0.5	0.5 to >0 pts	0 pts	
	Excellent	Good	Fair	Poor	
	Objectives are SMART and aligned with BSE methodology.	Objectives that pick only two of the “SMART” objectives	Objectives are weakly formulated.	Objectives missing or inappropriate.	
	Comments Where the objectives lose ground on the SMART standard is achievability. The second objective compresses an enormous amount of work into a single deliverable due by week seven: a PPO agent with a six-component reward function, a FastAPI backend, a Flask front end with four pages, a full SQLite schema, and a Unity 3D scene with ONNX inference, all built by one undergraduate in roughly five weeks after the environment is validated. Bundling that much scope into one objective makes it hard to tell whether the objective is really one achievable unit of work or the entire remaining project disguised as a single line, and it leaves only one week between a fully working system and a completed three-zone, three-seed evaluation in objective three, which is tight for debugging alone. The way forward is to either split objective two into two or three smaller objectives with their own checkpoints (agent and backend as one, front end and Unity visualisation as another), or to explicitly mark the Unity 3D demo and one or two of the four web pages as stretch goals outside the core SMART commitments, so that the achievability of the core research objectives isn't tied to the survival of the most time-consuming, lowest-research-value component.				0.85 / 1 pts
Significance and Scope [Content Scope, Geographical scope, and time Scope]	2 to >1.5 pts	1.5 to >1 pts	1 to >0 pts	0 pts	1.75 / 2 pts
	Mastery	Proficient	Developing	Fail	
	Strong justification and clearly defined content, geographical, and time scope.	Importance and scope are explained but need strengthening.	Weak rationale and unclear scope.	Significance and scope not adequately addressed.	

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Literature Review: Thematic Organization and Critical Analysis	Comments contains an internal contradiction that needs fixing. It states the project covers "nine terrain zones across all six agro-ecological regions" but then names only six regions: Northern Highlands, Central Plateau, Western Ridge, Eastern Savanna, Southern Valley, and Kivu Belt. Nine zones and six named regions cannot both be the complete picture without saying how the nine map onto the six, particularly since the project scope section earlier states six zones are used for training and three are held out for evaluation, which implies the nine-zone breakdown is the right one but it never gets reconciled with the six-region list given here.				
	2 to >1.5 pts Mastery Strong thematic organization and critical analysis.	1.5 to >1 pts Proficient Good organization but limited critical discussion.	1 to >0 pts Developing Weak organization and superficial review.	0 pts Fail No meaningful literature	
	Comments The structure is strong: a systems overview, three clear themes, a summary table, and a strengths/weaknesses section. Most sources are cited accurately, though the specific finding attributed to Dradoum et al. (2025), the LSTM-PPO comparison, isn't clearly supported by that paper's actual abstract and should be checked. The main problem is that the "analysis" mostly just names limitations without explaining why they exist. Each system in 2.2 gets the same treatment: describe it, then one sentence saying what it's missing. For example, AirSeed "does not address ecological placement at all" — true, but the document never says why: is it a hardware limit, a cost tradeoff, or just outside the system's original goal? The summary table has the same issue: the gaps listed are mostly what each paper already says about itself, not something drawn out by comparing sources to each other. No two prior works are ever weighed against one another directly; everything is compared only to ARIA. The two lines in 2.5, that commercial systems are "fast but ecologically static" and academic ones are "adaptive but not ecologically grounded," are the one place real synthesis happens. But it's a two-sentence conclusion after pages of description, not something built up gradually. To fix it: for each named limitation, add one sentence on the cause, not just the fact. And put at least a couple of sources in direct conversation with each other, not only with ARIA. That's the difference between describing the literature and analysing it.				1.5 / 2 pts
	2 to >1.5 pts Mastery	1.5 to >1 pts Proficient	1 to >0 pts Developing	0 pts Fail	1.5 / 2 pts

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Strategy	Appropriate and justified research design and sampling strategy.	Design and sampling described but weakly justified.	Inappropriate or vaguely described methodology.	Missing methodology.	
Comments First, the proposal never argues why reinforcement learning is the right family of method for this problem in the first place; section 3.4 justifies PPO over DQN and A2C, which is choosing among RL algorithms after RL has already been assumed, not defending RL against a non-learning alternative such as constrained optimisation over the same static, fully observable grid. Given that the environment is deterministic and modest in size, that comparison matters, and the one property that would clearly favour RL, a trained policy generalising instantly to new terrain zones without being re-solved, is never named as the reason for the choice. Second, the sampling strategy calls itself "stratified purposive sampling" but doesn't describe the actual selection procedure: how large the candidate pool was, what thresholds defined "highest ecological heterogeneity," or how nine zones were chosen over other combinations. With only three held-out zones and three random seeds per ablation condition, the proposal's later claim of assessing hypotheses "at the 95% confidence level" also needs a more careful statistical justification, since that is a small sample for a formal confidence claim. Naming the actual selection criteria and thresholds for the zone sampling, and adding a paragraph in this section that argues RL against a concrete non-learning alternative rather than only against other RL algorithms, would move this into the top band.					
Is the environment really deterministic or stochastic ?					
Ethical Considerations and References	1 to >0.75 pts	0.75 to >0.5	0.5 to >0 pts	0 pts	1 / 1 pts
	Mastery Ethical issues thoroughly identified and addressed.	Proficient Some ethical issues identified.	Developing Ethical concerns only partially addressed.	Fail Ethical considerations absent.	
Writing mechanics, formatting, and organization	1 to >0.75 pts	0.75 to >0.5	0.5 to >0 pts	0 pts	1 / 1 pts
	Mastery Clear, concise, organized, and	Proficient Minor writing and formatting	Developing Frequent writing and	Fail Poorly written and difficult to follow.	

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	error-free writing.	issues.	formatting issues.		
					
	1 to >0.75 pts	0.75 to >0.5 pts	0.5 to >0 pts	0 pts	
	Mastery	Proficient	Developing	Fail	
References	Relevant, credible, and correctly formatted references.	Minor citation or formatting errors.	Significant citation issues.	Missing or inappropriate references.	1 / 1 pts
					
Technical Diagrams	3 to >2 pts	2 to >1 pts	1 to >0 pts	0 pts	2.5 / 3 pts
view longer description	Mastery	Proficient	Developing	Fail	
	All required diagrams are present, correctly modeled, labeled, and aligned with the proposed system. Diagrams are professional, consistent, and provide a clear understanding of the solution.	All required diagrams are present but contain minor modeling or consistency issues.	Some required diagrams are missing or poorly constructed, with limited alignment to the proposed solution.	Diagrams are missing or largely incorrect.	
					
	Comments				
	The use case diagram (Figure 4) is where a real consistency issue shows up. The accompanying text states the Web User connects to six use cases, including "enforcing the protected area constraint" and the shared "View Agent Comparison Dashboard," but the diagram itself only shows the Web User linked to four ovals: selecting a zone, triggering an episode, viewing the seed placement map, and viewing the species guide. The "Enforce Protected Area Constraint" use case appears instead to be linked to the RL Agent and, via a crossing line, to the Evaluator, not to the Web User as the caption claims. This				

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	matters because the whole point of a use case diagram is to show, unambiguously, which actor triggers which behavior, and right now the written description and the actual drawn associations disagree on that point for two of the sixteen use cases. The fix is to either redraw the diagram so the Web User's associations match what the text claims, or rewrite the text to accurately describe what's actually drawn, since as it stands a reader can't tell which version is correct. Once that's reconciled, this criterion would sit comfortably in the Mastery band.		Total points: 15.6	
Final version of the product/solution Assignment 3: Final Capstone Implementation [S:A]	8 Jul by 23:59	8 Jul at 23:41		10 / 15 
Mission Capstone Exit Ticket: Compiling a Mission portfolio Imported Assignments	20 Jul by 23:59	20 Jul at 14:25		
Supervisor assessment of the students availability Assignment 5: Supervisor's grade [F:A]	20 Jul by 23:59	20 Jul at 22:09		 / 10
Capstone Report Submission Assignment 4: Capstone Report Submission [F:A]	24 Jul by 23:59			- / 35
Defense [Presentation and Demo] Assignment 6: Capstone Defense [S:A]	31 Jul by 23:59			 / 30
Final Submission Assignment 7: Final Capstone Report Submission [S:A]	31 Jul by 23:59			 / 5

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Roll Call Attendance Workshop Attendance [F:A]		29 May at 11:16		100% 
Assignments				100% 1.00 / 1.00
Assignment 1: Research proposal [F:A]				82.11% 15.60 / 19.00
Assignment 2: Initial Design and Development [F:A]				N/A 0.00 / 0.00
Assignment 3: Final Capstone Implementation [S:A]				66.67% 10.00 / 15.00
Assignment 4: Capstone Report Submission [F:A]				N/A 0.00 / 0.00
Workshop Attendance [F:A]				100% 100.00 / 100.00
Assignment 5: Supervisor's grade [F:A]				N/A 0.00 / 0.00
Assignment 6: Capstone Defense [S:A]				N/A 0.00 / 0.00
Assignment 7: Final Capstone Report Submission [S:A]				N/A 0.00 / 0.00
Imported Assignments				N/A 0.00 / 0.00
Total				78.46%

